

Mohammed Moniruzzaman Khan

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Professional Summary

Results-driven **Quantitative Analyst & Data Scientist** with expertise in **machine learning, deep learning, financial modeling, and data analytics**. Experienced in pricing and hedging financial derivatives, developing AI-driven predictive models, and deploying Python-based solutions. Skilled at combining advanced mathematics, stochastic modeling, and data engineering to deliver impactful solutions in finance, analytics, and ML-driven applications.

Core Competencies

Programming & Development: Python, MATLAB, SQL, Java, C

Machine Learning & AI: TensorFlow, Keras, Scikit-Learn, Deep Learning, Feature Engineering, Time Series Forecasting

Financial Modeling: PDEs, Stochastic Processes, Monte Carlo Simulations, Option Pricing

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, EDA, Statistical Testing

Data Engineering & Cloud: ETL, Spark, Docker, AWS (basic)

Optimization & Forecasting: Predictive Analytics, Model Optimization, Risk Management

Education

Nova School of Science & Technology, Lisbon

PhD in Mathematics

2024–Present

Thesis: *Future Research Problems: Pricing and Hedging Options with Machine Learning*

Key Courses: Computational Financial Mathematics, Stochastic Differential Equations, PDE Analysis

Baruch College, CUNY, USA

BA in Mathematics

1993–1999

Projects & Certifications

Customer Retention MLOps Pipeline: GitHub – Scalable pipeline predicting customer retention using classification models; deployed with Docker and AWS.

Financial Time Series Forecasting: GitHub – Forecasted SPY (S&P 500 ETF), QQQ (Nasdaq-100 ETF), and BTC (Bitcoin) prices using ARIMA, Random Forest, and LSTM models; feature engineering included moving averages, rolling volatility, and lagged returns.

American Options Pricing with Deep Learning: GitHub – Implemented Longstaff-Schwartz Monte Carlo (LSM) in MATLAB; neural networks enhanced hedging strategies and benchmarked against PDE & Monte Carlo methods.

Computational Finance PDE Solver: GitHub – Python-based finite difference solvers for European/American options, Monte Carlo & Black-Scholes modeling, deep learning integration for path-dependent derivatives.

Certifications: Kaggle: Intro to Machine Learning, Intro to SQL, Advanced SQL, Pandas, Feature Engineering, Data Cleaning, Intro to Deep Learning, Time Series Forecasting.

Le Wagon: Introduction to Docker (Free Online Certificate)

Professional Experience

The Stylex Collection, Dhaka, Bangladesh

IT In Charge

Sep 2018–Jan 2019

Managed IT infrastructure, Windows Server, network configuration, and security policies; improved system reliability and productivity through technical support and training.

IFIC Bank Ltd, Dhaka, Bangladesh

Senior Principal Officer

Jul 2013–Jul 2018

Administered SWIFT servers and customized Java-based MMS integrated with core banking; developed SQL Server components ensuring transactional data integrity.

IBCS-Primax Software Ltd, Dhaka, Bangladesh

Senior Programmer

Nov 2012–Jun 2013

Designed and optimized Java enterprise software modules; created Oracle 10g database schemas.

UTC Associates (Pvt) Ltd, Dhaka, Bangladesh

Software Engineer

Jan 2012–Oct 2012

Developed BI reports and optimized SQL queries for enterprise reporting.

ADN Technologies Ltd, Dhaka, Bangladesh

UI Architect / Project Lead

Apr 2011–Nov 2011

Modernized legacy POS system in C; integrated with back-office modules.

Empire Art Gallery, NY, USA

UI Architect / Software Consultant

1995–2009

Developed web/internal applications using HTML, CSS, JavaScript, SQL; improved inventory and e-commerce systems.

Achievements

- Designed and implemented machine learning and deep learning models for pricing and hedging American-style financial derivatives.
- Developed numerical PDE solvers and Monte Carlo simulation frameworks for option pricing with strong analytical accuracy and computational efficiency.
- Translated advanced mathematical theory (PDEs, SDEs, stochastic processes) into reproducible, well-documented Python and MATLAB implementations.
- Built and maintained a strong open-source GitHub portfolio demonstrating end-to-end quantitative finance, machine learning, and data science workflows.

Languages

English: Fluent

Bengali: Native

References

Marta Faias: PhD Supervisor – mcm@fct.unl.pt

Pedro Mota: PhD Supervisor – pjpm@fct.unl.pt